

# Onur Tellioglu

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github.com/onur-tellioglu

## Summary

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BSc AI student at FAU, working as a research assistant in the university's photonics lab. Writing production code since 2021. Recent work spans instrument control and acquisition software for lasers, spectrometers and RF hardware, LLM agent systems in production (a WhatsApp system running the complete patient journey for a surgical clinic), a turbine engine controller with a full sensor suite, and a self-hosted webhook relay written in Go.

## Experience

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**Student Research Assistant (HiWi)**, FAU, Institute of Microwaves and Photonics (LHFT) June 2026 – present

- Wrote the Python control library for a Luna/Polytec Phoenix 1400 tunable laser. The vendor ships a 32-bit DLL with C++, LabVIEW and MATLAB examples and no Python binding, so this one loads it through ctypes behind a class that validates values and tracks laser state. Delivered in four rounds, each signed off on the real instrument.
- Built custom acquisition and analysis software for an Ibsen I-MON 256 USB fiber Bragg grating interrogator, replacing the vendor evaluation tool: high-speed spectrum capture (~6000 frames/s over FTDI FT232H), peak tracking and calibration.
- Drove an AD9910 DDS evaluation board over SPI from a Raspberry Pi Pico instead of the vendor USB stack. Registers read back exact and the DAC tracks the programmed tuning word to within 1 ppm.
- Rebuilt the lab's GPIB stack on Ubuntu 24.04 as a one-shot installer (linux-gpib pinned at r2107, compat bridge for the kernel 6.16+ timer API) with an update freeze so kernel upgrades cannot break it.

**Co-Founder & Lead Developer**, Assistbase – Remote Feb 2025 – Jan 2026

- Built a fully custom system running the complete patient journey over WhatsApp for a surgical clinic, with no low-code tools or templates: multilingual intake, photo and health data collection, booking, reminders, post-op follow-up. Handles 90% of communication end-to-end and escalates to staff only when needed.
- Built a multi-dimensional patient memory system: every conversation is parsed into a structured profile (medical history, procedure interests, budget signals, travel context) persisted in Supabase, giving the agent stateful behaviour across sessions.
- Delivered a separate paperless operations platform for a manufacturing facility in Germany: raw material intake, production records, deliveries with digital signatures, dual-layer finance, i18n (TR/DE). Live in production, SaaS expansion planned. Stack: Node.js, OpenAI API, Supabase, Next.js, Prisma, PostgreSQL.

**Chief IT Consultant**, Metarc Interior – Istanbul July 2023 – Oct 2023

- Rebuilt the full IT infrastructure of a corporate architecture firm: secure offsite backup, office telephony, cybersecurity hardening.

**Software Developer Intern**, HEFA Technology – Istanbul Aug 2022 – Sept 2022

- Contributed to airpm.io, an air quality monitoring platform. Integrated IoT sensors via LoRa-WAN and The Things Network, built WebHooks for data ingestion, and created visualisation pipelines (.NET + MongoDB).

**Software Developer Intern**, Naylalabs – Istanbul Apr 2022 – May 2022

- Backend development in Node.js: REST APIs, CLI tooling with Yargs, third-party API integrations.

**Technical Writer & Web Administrator**, ModArt PC – Istanbul Dec 2020 – June 2023

- Wrote SEO-optimized tech news, buying guides, and in-depth product reviews for a PC hardware media outlet, with hands-on benchmark testing. Managed web infrastructure: DNS, SSL, Cloudflare, security hardening.

## Education

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**Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)**, B.Sc. in Artificial Intelligence Oct 2025 – present

- GPA: 1.4 (German scale)

**Istanbul Arel University**, B.Sc. in Computer Engineering

Sept 2023 – June 2024

- GPA: 3.43 / 4.00, transferred to FAU

**Istanbul Aydın University**, Associate Degree in Computer Programming

Sept 2020 – June 2022

- GPA: 3.24 / 4.00, Honor Student

## Projects

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### **PomeloHook: Self-Hosted Webhook Relay**

Apr 2026 – present

- [hook.pomelostudios.net](https://hook.pomelostudios.net): think ngrok, but self-hosted, team-aware, and built around persistent event history.
- Go server + CLI binary with an embedded React/Vite dashboard. Events land in SQLite before forwarding over the WebSocket tunnel, so they stay replayable from CLI or dashboard regardless of forwarding outcome. Personal and org tunnels, 30-day retention, single binary deployment.

### **turbin-ecu: Micro-Turbine Engine Controller**

Mar 2026

- ESP32 engine control unit driven by a 9-state FSM (IDLE → PRE\_HEAT → IGNITION → WARMUP → RAMP\_UP → RUNNING → SHUTDOWN → COOLING).
- Interrupt-driven Hall-effect RPM, MAX31855 thermocouple over SPI, throttle ADC; MOSFET-driven solenoid valves, brushless ESC and fuel pump on PWM, plus desktop simulation via the PlatformIO native target. Stack: C++, PlatformIO, ESP32.

### **ctscan & mend: macOS System Health**

Apr 2026 – present

- [onur-tellioglu.github.io/ctscan](https://onur-tellioglu.github.io/ctscan): a 14-module CLI published as a Homebrew formula, pure Bash and dependency-free: battery, SSD wear (S.M.A.R.T.), thermals, memory, Wi-Fi, FileVault/SIP/Gatekeeper.
- mend is the SwiftUI macOS app wrapping that engine via the Process API, with menu bar integration and a background scan timer. The commercial app and the open-source tool are intentionally separate.

### **Cull: Swipe-to-Cull iOS Photo Manager**

Jun 2026

- SwiftUI app that reviews a photo library one image at a time, learns what is blurry, and keeps, culls or undoes with a single gesture. Submitted to App Store review.

## Side Projects

- [FAUVault.space](https://FAUVault.space): study platform for FAU students; 60%+ of one semester's exam directly covered by the content, with measurable adoption across courses.
- [kampusdizi.vercel.app](https://kampusdizi.vercel.app) and [stan-quiz.vercel.app](https://stan-quiz.vercel.app): Next.js side sites. Browser multiplayer games on Phaser and Colyseus.

## Skills

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**LLM & Agents:** Claude API, OpenAI API, agent orchestration, tool/function calling, retrieval-augmented generation, semantic search, custom memory systems

**Embedded & IoT:** ESP32, Raspberry Pi Pico, Arduino, C++ (PlatformIO), SPI/UART/ADC, DDS (AD9910), GPIB/linux-gpib, ctypes instrument bindings, LoRa-WAN, TTN, MOSFET drivers, FSM design, OpenSCAD

**Programming Languages:** Python, TypeScript/JavaScript, Go, C#, Swift, C++, Java, SQL, Bash

**Web & Mobile:** Node.js, Next.js, React, React Native (Expo), SwiftUI, WidgetKit, Supabase, Prisma, PostgreSQL, MongoDB, Firebase, Cloudflare, Vite, Electron.js

**Engineering Practice:** TDD, CI/CD (GitHub Actions), Docker, pnpm monorepos, row-level security, i18n, design systems

**Spoken Languages:** Turkish (native), English (fluent), German (B1)